

TAMIL NADU STATE BOARD - STANDARD 11 MATHEMATICS VOLUME 1

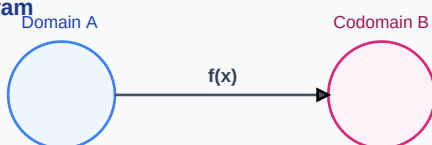
CHAPTER 4: COMBINATORICS AND MATHEMATICAL INDUCTION

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 11 English Medium
Subject:	Mathematics Volume 1
Chapter Name:	Combinatorics and Mathematical Induction

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Function Mapping Diagram

Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Permutation	An ordered arrangement of a set of objects, where the order of selection is highly significant.
Combination	A selection of a set of objects without regard to order, focusing only on the grouping of items.
Principle of Mathematical Induction	A proof technique where a statement is shown to hold for $n=1$, assumed for $n=k$, and proved for $n=k+1$.

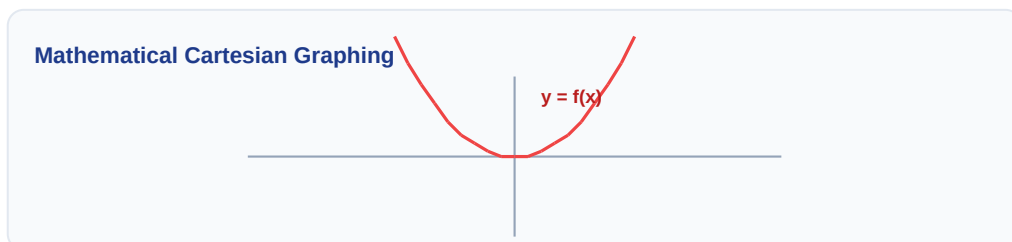
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

Venn Diagram: $A \cap B$



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



Essential Formulas, Laws & Identities:

- Permutations: $nPr = \frac{n!}{(n-r)!}$
- Combinations: $nCr = \frac{n!}{r! (n-r)!}$
- Addition Principle: Total ways = $m + n$ (for mutually exclusive events)
- Multiplication Principle: Total ways = $m * n$ (for successive events)

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Find the value of n if $nP_4 = 12 * nP_2$.

Step-by-step Solution:

$$nP_4 = n(n-1)(n-2)(n-3) \text{ and } nP_2 = n(n-1).$$

$$\text{Given: } n(n-1)(n-2)(n-3) = 12 * n(n-1).$$

$$\text{For } n > 3: (n-2)(n-3) = 12 \Rightarrow n^2 - 5n + 6 = 12 \Rightarrow n^2 - 5n - 6 = 0 \Rightarrow (n-6)(n+1) = 0.$$

Since n must be positive, $n = 6$.

Linear Algebraic Matrix Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Prove by mathematical induction that $1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$.

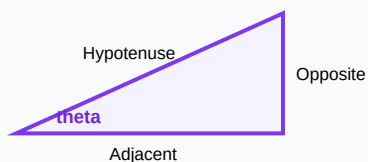
Analytical Solution Strategy:

Base case $n=1$: LHS = 1, RHS = $\frac{1(2)(3)}{6} = 1$. Assume true for $n=k$: $S_k = \frac{k(k+1)(2k+1)}{6}$.

For $n=k+1$: $S_{(k+1)} = S_k + (k+1)^2 = \frac{k(k+1)(2k+1)}{6} + (k+1)^2$.

Factor out $(k+1)$: $(k+1)[\frac{k(2k+1)}{6} + k+1] = \frac{(k+1)}{6}[2k^2 + 7k + 6] = \frac{(k+1)(k+2)(2k+3)}{6}$. Proved.

Right-Angled Triangle Geometry



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Find the number of 4-letter words that can be formed using letters of 'PENCIL'.

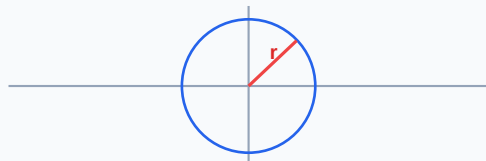
Q2 (2-Mark): Prove that $nCr + nC(r-1) = (n+1)Cr$.

Q3 (2-Mark): If $nC8 = nC2$, find the value of $nC21$.

Q4 (5-Mark): In how many ways can 5 people be seated around a circular table?

Q5 (5-Mark): Use mathematical induction to prove that $7^n - 3^n$ is divisible by 4.

Circle with Radius Vector on Plane



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Cryptography

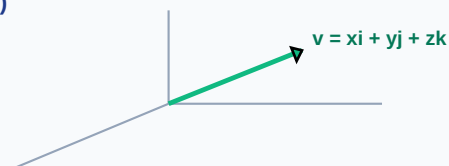
Password complexities are evaluated using permutation formulas to determine vulnerability sizes.

Application Case: Logistics

Delivery routes are optimized using counting trees to find the shortest delivery paths.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

3D Vector Projections (x, y, z)



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Probability Tree Outcome Diagram

